5.2 TORQUE AND ANGULAR MOMENTUM

When a net force acts on a body, it produces linear motion in the direction of the applied force. If the body is fixed to a point or an axis, such a force rotates the body depending on the point of application of the force on the body. This ability of the force to produce rotational motion in a body is called torque or moment of force. Examples for such motion are plenty in day to day life. To mention a few; the opening and closing of a door about the hinges and turning of a nut using a wrench.



The extent of the rotation depends on the magnitude of the force, its direction and the distance between the fixed point and the point of application. When torque produces rotational motion in a body, its angular momentum changes with respect to time. In this Section we will learn about the torque and its effect on rigid bodies.

5.2.1 Definition of Torque

Torque is defined as the moment of the external applied force about a point or axis of rotation. The expression for torque is,

$$\vec{\tau} = \vec{r} \times \vec{F} \qquad (5.9)$$

where, $\vec{r}$ is the position vector of the point where the force $\vec{F}$ is acting on the body as shown in Figure 5.4.

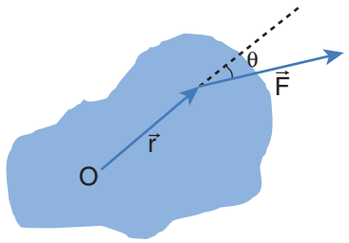


Figure 5.4 Torque on a rigid body

Here, the product of $\vec{r}$ and $\vec{F}$ is called the *vector product or cross product*. The vector product of two vectors results in another vector that is perpendicular to both the vectors (refer Section 2.5.2). Hence, torque $(\vec{\tau})$ is a vector quantity.

Torque has a magnitude $rF\sin\theta$ and direction perpendicular to $\vec{r}$ and $\vec{F}$. Its unit is N m.

$$\vec{\tau} = (rF\sin\theta)\hat{n} \qquad (5.10)$$

Here, θ is the angle between $\vec{r}$ and $\vec{F}$, and $\hat{n}$ is the unit vector in the direction of $\vec{\tau}$. Torque $(\vec{\tau})$ is sometimes called as a *pseudo vector* as it needs the other two vectors $\vec{r}$ and $\vec{F}$ for its existence.

The direction of torque is found using right hand rule. This rule says that if fingers of right hand are kept along the position vector with palm facing the direction of the force and when the fingers are curled the thumb points to the direction of the torque. This is shown in Figure 5.5.

The direction of torque helps us to find the type of rotation caused by the torque. For example, if the direction of torque is out

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